National Academy of Science and Technology

Affiliated to Pokhara University

Accredited by University Grants Commission (UGC), Nepal 2022.

Uttar behadi-4, Dhangadhi, Kailali

A

Lab Report

Of

"Object-Oriented Programming Using Java"

For the partial fulfillment of degree of Computer Application

Awarded by Pokhara University

Submitted to

Department of Computer Application

National Academy of Science and Technology

Under the supervision of

Mr. Sunil Bahadur Bist

Lecturer, Department of Computer Application

Submitted by

Gajendra Awasthi (24530017)

Feb 2026

Submission Date

Signature

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Basic Structure Of Java Program

Program:

```
class JavaStructure {
	public static void main(String[] args) {
		System.out.println("   ");
		System.out.println("Basic Java Program Structure");
		System.out.println("   ");
		displayMessage();
		System.out.println("Program executed successfully");
	}
	static void displayMessage() {
		System.out.println("This is java program");
	}
}
```

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Output:

```
Basic Java Program Structure

This is java program
Program executed successfully


...Program finished with exit code 0
Press ENTER to exit console.
```

Datatypes Variables & Operators

Program:

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```
class DataTpOperators {
        int x = 30;
        int y = 50;

        void arithmeticOperations() {
                System.out.println("Addition: " + (x + y));
                System.out.println("Subtraction: " + (x - y));
                System.out.println("Multiplication: " + (x * y));
                System.out.println("Division: " + (x / y));
        }

        void relationalOperations() {
                System.out.println("x > y: " + (x > y));
                System.out.println("x < y: " + (x < y));
                System.out.println("x == y: " + (x == y));
        }

        public static void main(String[] args) {
                DataTpOperators obj = new DataTpOperators();
                obj.arithmeticOperations();
                obj.relationalOperations();
        }
}
```

Output:

```
Addition: 80
Subtraction: -20
Multiplication: 1500
Division: 0
x > y: false
x < y: true
x == y: false
```

Java Program: Input & Output of Primitive Data Types

Code:

```
import java.util.Scanner;
public class PrimitiveInputDemo {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter integer value: ");
      int i = sc.nextInt();
      System.out.print("Enter float value: ");
      float f = sc.nextFloat();
      System.out.print("Enter double value: ");
      double d = sc.nextDouble();
      System.out.print("Enter character value: ");
      char c = sc.next().charAt(0);
      System.out.print("Enter boolean value: ");
      boolean b = sc.nextBoolean()
      System.out.println("\n--- Output ---");
      System.out.println("Integer  : " + i);
      System.out.println("Float     : " + f);
      System.out.println("Double  : " + d);
      System.out.println("Character: " + c);
      System.out.println("Boolean  : " + b);
      sc.close();
   }
}
```

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Output:

```
Enter integer value: 2
Enter float value: 4.4
Enter double value: 14.4
Enter character value: G
Enter boolean value: true

--- Output ---
Integer : 2
Float   : 4.4
Double : 14.4
Character: G
Boolean : true
```

1. Abstract Class Example

Code:

```
abstract class Animal {
    abstract void sound();
    void eat() {
        System.out.println("Animal is eating");
    }
}
class Dog extends Animal {
    void sound() {
        System.out.println("Dog barks");
    }
}
public class AbstractDemo {
    public static void main(String[] args) {

        Animal a = new Dog();
        a.sound();
        a.eat();
    }
}
```

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Output:

```
Dog barks
Animal is eating


...Program finished with exit code 0
Press ENTER to exit console.
```

2. Arrays, Strings & Control Structure

Program:

```
class ArrayStringControl {
	void arrayExample() {
		int[] marks = {70, 80, 80, 90, 99};
		int sum = 0;

		for (int i = 0; i < marks.length; i++) {
			sum += marks[i];
		}

		System.out.println("Total Marks: " + sum);
		System.out.println("Average Marks: " + (sum / marks.length));
	}

	void stringExample() {
		String course = "Java Object Oriented Programming";
		System.out.println("Course Name: " + course);
		System.out.println("Length: " + course.length());
	}

	public static void main(String[] args) {
		ArrayStringControl obj = new ArrayStringControl();
		obj.arrayExample();
		obj.stringExample();
	}
}
```

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Output:

```
Total Marks: 419
Average Marks: 83
Course Name: Java Object Oriented Programming
Length: 32


...Program finished with exit code 0
Press ENTER to exit console.
```

3. Class & Constructors

Program:

```
class StudentConstructor {
        int id;
        String name;
        StudentConstructor() {
                id = 1;
                name = "Funsuk Wangdu";
        }
        StudentConstructor(int i, String n) {
                id = i;
                name = n;
        }
        void display() {
                System.out.println("ID: " + id);
                System.out.println("Name: " + name);
        }
        public static void main(String[] args) {
                StudentConstructor s1 = new StudentConstructor();
                StudentConstructor s2 = new StudentConstructor(2, "Raju Rastogi");

                s1.display();
                s2.display();
        }
}
```

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Output:

```
ID: 1
Name: Funsuk Wangdu
ID: 2
Name: Raju Rastogi


...Program finished with exit code 0
Press ENTER to exit console.
```

4. Method Overloading (class & Object)

Program:

```
class MethodOverloading {

int add(int a, int b) {
    return a + b;
}

int add(int a, int b, int c) {
    return a + b + c;
}

double add(double a, double b) {
    return a + b;
}

public static void main(String[] args) {
    MethodOverloading obj = new MethodOverloading();

    System.out.println(obj.add(10, 20));
    System.out.println(obj.add(10, 20, 30));
    System.out.println(obj.add(5.5, 6.5));
}

}
```

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Output:

```
60
100
17.49


...Program finished with exit code 0
Press ENTER to exit console.
```

5. Inheritance & Method Overloading

Code:

```
class MethodOverloading {
    // Method to add two integers
    int add(int a, int b) {
        return a + b;
    }

    // Method to add three integers
    int add(int a, int b, int c) {
        return a + b + c;
    }

    // Method to add two doubles
    double add(double a, double b) {
        return a + b;
    }

    public static void main(String[] args) {
        MethodOverloading obj = new MethodOverloading();
        System.out.println(obj.add(70, 50));
        System.out.println(obj.add(10, 50, 70));
        System.out.println(obj.add(50.5, 6.50));
    }
}
```

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Output:

```
120
130
57.0

...Program finished with exit code 0
Press ENTER to exit console.
```

6. Packages

Creating a Package

a. **Package Class: Student.java**

Path: mypackage.java

```
package mypack; public class Hello {
public void show() {
System.out.println("Hello from package!");
}
}
```

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b. **Main Class: MainApp.java**

Path: MainApp.java

```
import mypack.Hello; public class Main {
public static void main(String[] args) {
Hello obj = new Hello();
obj.show();
}
}
```

```
C:\users\utgs1\Desktop\cybgaz> javac -d . Hello.java
C:\users\utgs1\Desktop\cybgaz> java Main
Hello from package!
```

7. **Write a program to create an interface account and 3 abstract methods deposit (double) withdrawal and getbalance and derived two classes, is saving, one is saving account and another one is current account and implement it.**

Code:

```
interface Account {
	void deposit(double amount);
	void withdraw(double amount);
	double getBalance();
}

// Saving Account class
class SavingAccount implements Account {
	private double balance = 0;
	public void deposit(double amount) {
		balance += amount;
	}
	public void withdraw(double amount) {
		if (amount <= balance) balance -= amount;
		else
			System.out.println("Insufficient balance in Saving Account");
	}
	public double getBalance() {
		return balance;
	}
}

// Current Account class
class CurrentAccount implements Account {
	private double balance = 0;
	public void deposit(double amount) {
		balance += amount;
	}


	public void withdraw(double amount) {

		balance -= amount;  // overdraft allowed
	}
	public double getBalance() {
		return balance;
	}
}

public class InterfaceDemo {
	public static void main(String[] args) {
		Account sa = new SavingAccount();
		sa.deposit(7000);
```

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```
            sa.withdraw(5000);
            System.out.println("Saving Account Balance: " + sa.getBalance());

            Account ca = new CurrentAccount();
            ca.deposit(8000);
            ca.withdraw(2000);
            System.out.println("Current Account Balance: " + ca.getBalance());
        }
}
```

Output:

```
Saving Account Balance: 2000.0
Current Account Balance: 6000.0


...Program finished with exit code 0
Press ENTER to exit console.
```

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8. Exception Handling

Types of Exceptions with Examples

a. Checked Exceptions (Compile-time Exceptions)

i. IOException in File Handling

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```
import java.io.*;
class FileReadDemo {
        public static void main(String[] args) {
                try {
                        FileReader fr = new FileReader("data.txt");
                        int ch;
                        while ((ch = fr.read()) != -1) {
                                System.out.print((char) ch);
                        }
                        fr.close();
                } catch (IOException e) {
                        System.out.println("File not found or error reading file");
                }
        }
}
```

Output:

```
File not found or error reading file

...Program finished with exit code 0
Press ENTER to exit console.
```

b. Unchecked Exceptions (Runtime Exceptions)

i. ArithmeticException

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```
class ArithmeticDemo {
public static void main(String[] args) {
        try {
                int a = 10, b = 0;
                int c = a / b;
                System.out.println(c);
        } catch (ArithmeticException e) {
                System.out.println("Division by zero error handled");
        } finally {
                System.out.println("Finally block executed");
        }
    }
}
```

Output:

```
Division by zero error handled
Finally block executed


...Program finished with exit code 0
Press ENTER to exit console.
```

ii. ArrayIndexOutOfBoundsException

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```
class ArrayDemo {
        public static void main(String[] args) {
                int[] arr = {1, 2, 3};
                try {
                        System.out.println(arr[5]); // Invalid index
                } catch (ArrayIndexOutOfBoundsException e) {
                        System.out.println("Array index out of bounds enter less data");
                }
        }
}
```

Output:

```
Array index out of bounds enter less data

...Program finished with exit code 0
Press ENTER to exit console.
```

iii. NullPointerException

```
class NullPtDemo {
        public static void main(String[] args) {
                String str = null;
                try {
                        System.out.println(str.length());
                } catch (NullPointerException e) {
                        System.out.println("Null value cannot be used enter data");
                }
        }
}
```

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Output:

```
Null value cannot be used enter data

...Program finished with exit code 0
Press ENTER to exit console.
```

iv. NumberFormatException

```
class NumberFormatExcep {
    public static void main(String[] args) {
        String str = "abc123";
        try {
            int num = Integer.parseInt(str);
        } catch (NumberFormatException e) {
            System.out.println("Invalid number format");
        }
    }
}
```

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Output:

```
Invalid number format

...Program finished with exit code 0
Press ENTER to exit console.
```

9. **WAP to write a Java program to copy and paste an audio, video, or image file using byte stream input and output operations.**

Program:

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```
import java.io.*;
class DataCopyMedia
{
	public static void main(String[] args)
	{
		try{
			int b;
			FileInputStream a1 = new FileInputStream("audio.mp3");
			FileOutputStream a2 = new FileOutputStream("audio_copy.mp3");
			while ((b = a1.read()) != -1) a2.write(b);

			a1.close();
			a2.close();
			FileInputStream v1 = new FileInputStream("video.mp4");
			FileOutputStream v2 = new FileOutputStream("video_copy.mp4");
			while ((b = v1.read()) != -1) v2.write(b);

			v1.close();
			v2.close();
			FileInputStream i1 = new FileInputStream("image.jpg");
			FileOutputStream i2 = new FileOutputStream("image_copy.jpg");
			while ((b = i1.read()) != -1) i2.write(b);

			i1.close();
			i2.close();
			System.out.println("Audio, Video and Image copied successfully");
		}
		catch (Exception e){
			System.out.println("Error occurred");
		}
	}
}
```

Output:

```
Error occurred

...Program finished with exit code 0
Press ENTER to exit console.
```

10. Multithreading

Program:

```
class FirstThread extends Thread{
        public void run(){
                for (int i = 1; i <= 5; i++){
                        System.out.println("Thread 1st: " + i);
                }
        }
}
class SecondThread extends Thread{
        public void run(){
                for (int i = 1; i <= 5; i++)
                {
                        System.out.println("Thread 2nd: " + i);
                }
        }
}

class MultiThreadDemo{
        public static void main(String[] args) {
                FirstThread t1 = new FirstThread();
                SecondThread t2 = new SecondThread();
                t1.start();
                t2.start();
        }
}
```

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Output:

```
Thread 1st: 3
Thread 2nd: 5
Thread 1st: 4
Thread 1st: 5


...Program finished with exit code 0
Press ENTER to exit console.
```

11. I/O Operations JDBC & JAVA FX

- FILE INPUT / OUTPUT

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```
import java.io.*;
class FileIODemo {
    public static void main(String[] args) throws IOException {
        FileWriter fw = new FileWriter("data.txt");
        fw.write("Java File Handling Is Started");
        fw.close();
        FileReader fr = new FileReader("data.txt");
        int cha;
        while ((cha = fr.read()) != -1) {
            System.out.print((char) cha);
        }
        fr.close();
    }
}
}
```

Output:

```
Java File Handling Is Started

...Program finished with exit code 0
Press ENTER to exit console.
```

12.Program to read & write 16-bit data

```
import java.io.*;
public class FileRW {
        public static void main(String[] args) throws IOException {


                FileWriter fw = new FileWriter("charFile.txt");
                fw.write("Hello Java");
                fw.close();


                FileReader fr = new FileReader("charFile.txt");
                int ch;

                System.out.println("Reading from file:");
                while ((ch = fr.read()) != -1) {
                        System.out.print((char) ch);
                }
                fr.close();
        }
}
```

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Output:

```
Reading from file:
Hello Java

...Program finished with exit code 0
Press ENTER to exit console.
```

13.Program to read & write 8-bit data

```
import java.io.*;
public class FileInputOutputStreamDemo {
        public static void main(String[] args) throws IOException {


                FileOutputStream fos = new FileOutputStream("byteFile.txt");
                String data = "Hello Java";
                fos.write(data.getBytes());
                fos.close();

                FileInputStream fis = new FileInputStream("byteFile.txt");
                int b;
                System.out.println("Reading from file:");
                while ((b = fis.read()) != -1) {
                        System.out.print((char) b);
                }
                fis.close();
        }
}
```

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Output:

```
Reading from file:
Hello Java

...Program finished with exit code 0
Press ENTER to exit console.
```

14.JDBC Program

```
import java.sql.*;

class JDBCDemo {

        public static void main(String[] args) {

                try {

Class.forName("com.mysql.cj.jdbc.Driver");

Connection con = DriverManager.getConnection( "jdbc:mysql://localhost:3306/testdb","root",
"password");

 Statement stmt = con.createStatement();

 ResultSet rs = stmt.executeQuery("SELECT * FROM student");

        while (rs.next()) {

        System.out.println(rs.getInt(1) + " " + rs.getString(2));

      }

                con.close();

                } catch (Exception e) {

                                System.out.println(e);

                }

        }

}
```

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Output:

```
C:\users\utgs1\.jdk-25.0.2\bin\j
ID      Name
----------------
1      Gajendra
2         Bibhu
3          Ajju
4         Kamal

Process finished with exit code 0
```

15.Authenticate a User from Database

Code:

```
import java.sql.*;
import java.util.Scanner;
public class UserAuthentication {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter username: ");
        String username = sc.nextLine();

        System.out.print("Enter password: ");
        String password = sc.nextLine();

        try {
            // Load MySQL Driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Connect to database
            Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/college", "root", "password");

            // Prepare query
            String sql = "SELECT * FROM users WHERE username=? AND password=?";
            PreparedStatement pst = con.prepareStatement(sql);
            pst.setString(1, username);
            pst.setString(2, password);

            ResultSet rs = pst.executeQuery();

            if (rs.next()) {
                System.out.println("Login successful! Welcome, " + username);
            } else {
                System.out.println("Invalid username or password");
            }
```

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```
            con.close();
        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }

        sc.close();
    }
}
```

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Output:

```
Enter username: Gajendra
Enter password(can leave empty): 12345
Login Successful! Welcome, Gajendra
```

16.Count Number of Students When Program is Supplied by User Code:

```
import java.sql.*;
public class CountStudentsByCourse {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/college";
        String user = "root";
        String password = "password";  // change this

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            String sql =
                "SELECT course, COUNT(*) AS total_students " +
                "FROM students GROUP BY course";

            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery(sql);

            System.out.println("Course-wise Student Count:");
            System.out.println("-------------------------- ");

            while (rs.next()) {
            String course = rs.getString("course"); int
            count = rs.getInt("total_students");
                System.out.println(course + " : " + count);
            }
            con.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

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Output:

```
C:\users\utgs1\.jdk-25.0.2\bin\j
Course-wise Student List:
---------------------------
Course: BCA
Students:
- Gajendra Awasthi
- Ajju
- Bibhu
- kamal
- Hemant
Total Students in BCA: 5

Course: BE
Students:
- Raju Rastogi
- Funsuk Wangdu
- Kumar Lal
- Mr Virus
Total Students in BE : 4

Process finished with exit code 0
```

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17.JAVA FX Program (Exam Oriented)

Drawing Line, Rectangle, and Text

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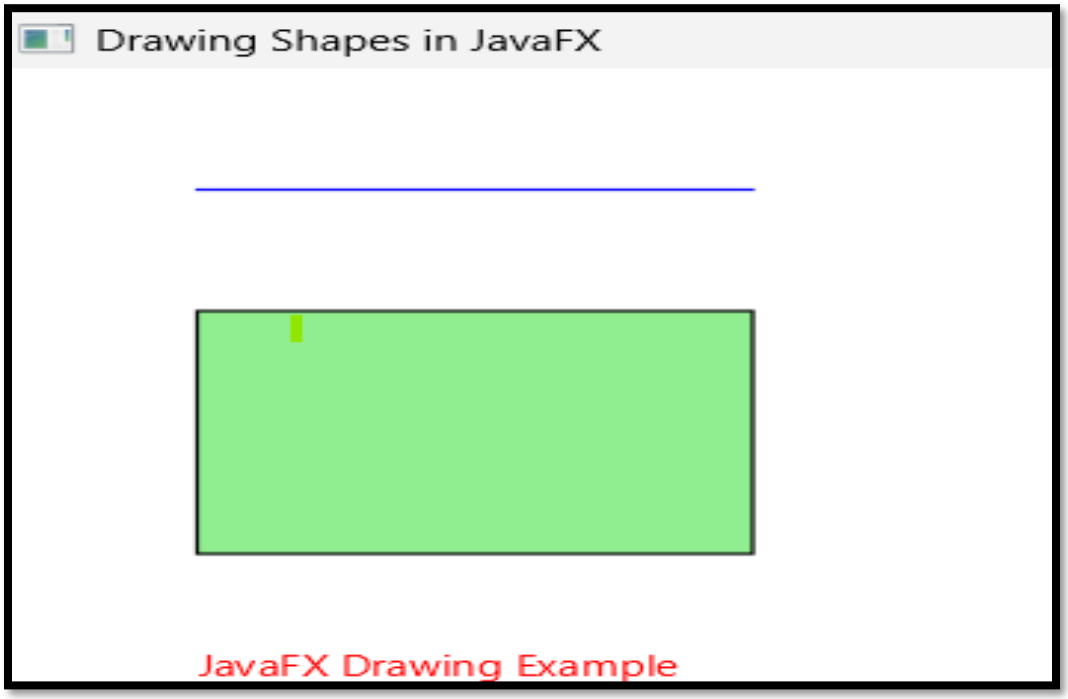
Code:

```
import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.layout.Pane;
import javafx.scene.paint.Color;
import javafx.scene.shape.Line;
import javafx.scene.shape.Rectangle;
import javafx.scene.text.Text;
import javafx.stage.Stage;

public class DrawingDemo extends Application {
    @Override
    public void start(Stage stage) {
        Pane pane = new Pane();
        // Draw a line
        Line line = new Line(50, 50, 200, 50);
        line.setStroke(Color.BLUE);
        // Draw a rectangle
        Rectangle rect = new Rectangle(50, 100, 150, 100);
        rect.setFill(Color.LIGHTGREEN);
        rect.setStroke(Color.BLACK);
        // Add text
        Text text = new Text(50, 250, "JavaFX Drawing Example");
        text.setFill(Color.RED);
        pane.getChildren().addAll(line, rect, text);
        Scene scene = new Scene(pane, 400, 300);
        stage.setTitle("Drawing Shapes in JavaFX");
        stage.setScene(scene);
        stage.show();
    }
    public static void main(String[] args) {
        launch(args);
    }
}
```

Output:



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18.Login & Signup Form with TextField and PasswordField

Code:

```
import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.control.Label;
import javafx.scene.control.PasswordField;
import javafx.scene.control.TextField;
import javafx.scene.layout.GridPane;
import javafx.stage.Stage;
public class LoginSignupDemo extends Application {
   @Override
   public void start(Stage stage) {
      GridPane grid = new GridPane();
      grid.setVgap(10);
      grid.setHgap(10);
      // Username
      Label userLabel = new Label("Username:");
      TextField userField = new TextField();
      grid.add(userLabel, 0, 0);
      grid.add(userField, 1, 0);
      // Password
      Label passLabel = new Label("Password:");
      PasswordField passField = new PasswordField();
      grid.add(passLabel, 0, 1);
      grid.add(passField, 1, 1);
      // Info label
      Label infoLabel = new Label();
      grid.add(infoLabel, 1, 3);
      // Login button
      Button loginBtn = new Button("Login");
      loginBtn.setOnAction(e -> {
      infoLabel.setText("Username: " + userField.getText() + ", Password: " + passField.ge tText());
      });
      grid.add(loginBtn, 1, 2);
      Scene scene = new Scene(grid, 400, 200);
      stage.setTitle("Login Form Example");
      stage.setScene(scene);
      stage.show();}
   public static void main(String[] args) {
      launch(args);
   }}
```

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Output:

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